25901797 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATIONS, APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCSAC100 - MUSCULOSKELETAL SYSTEM AND BIOMECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What is the main function of smooth muscle tissue? [U] / [CO1]

   a). Movement of limbs | b). Contraction of the heart | c). Regulation of internal organ movements | d). Protection of internal organs

2. Which type of cartilage is found in the joints to reduce friction? [U] / [CO2]

   a). Hyaline cartilage | b). Fibrocartilage | c). Elastic cartilage | d). Cartilage tissue is not found in joints

3. In a typical first-class lever system in the body (like the head and neck), where is the fulcrum located? [C] / [CO5]

   a). Between the load and the effort | b). At the distal end of the lever

   c). Between the load and the muscle | d). At the point where the muscle is attached to the bone

4. Which of the following factors affects the joint angular velocity during movement? [U] / [CO6]

a). Muscle fiber type

b). Muscle contraction velocity

c). Muscle cross-sectional area

d). Muscle length at the time of contraction

5. What type of joint is found at the elbow? [U] / [CO6]

a). Hinge joint

b). Ball and socket joint

c). Pivot joint

d). Saddle joint

6. According to the size principle, which motor units are recruited first during a low-intensity contraction? [U] / [CO6]

a). Large, fast-twitch motor units

b). Small, slow-twitch motor units

c). Large, slow-twitch motor units

d). Both large and small motor units are recruited simultaneously.

7. What is the effect of a muscle's moment arm (distance from the axis of rotation) on the efficiency of force generation in the musculoskeletal system? [C] / [CO5]

a). A longer moment arm allows for more efficient force generation at the joint.

b). A longer moment arm reduces the ability to generate force.

c). The moment arm has no effect on force generation.

d). A longer moment arm decreases the speed of movement.

8. Which biomechanical principle helps to explain how a small force can move a large load with the help of a lever system? [U] / [CO4]

a). Principle of work and energy

b). Principle of mechanical advantage

c). Principle of equilibrium

d). Principle of inertia

9. Which muscle is primarily supplied by the brachial artery and innervated by the radial nerve, which can be observed on a cadaveric specimen? [An] / [CO3]

a). Triceps brachii

b). Biceps brachii

c). Flexor carpi radialis

d). Flexor digitorum superficialis

10. Which of the following muscle types is under voluntary control? [U] / [CO1]

a). Skeletal muscle b). Smooth muscle

c). Cardiac muscle d). All of the above

[1x10 = 10]

Part B

Short Answer

Answer any **10** questions

Each question carries **2** marks

11. Name the four muscles in the quadriceps group. [U] / [CO1]

12. Identify two muscles responsible for neck movement. [U] / [CO1]

13. Define isotonic concentric contraction with an example. [K] / [CO2]

14. How do tendons help in movement? [K] / [CO3]

15. What are the main functions of the skeletal system? [U] / [CO3]

16. Describe the length-tension relationship in muscles and relate it to optimal force production. [U] / [CO1, CO2]

17. Evaluate the role biomechanics plays in enhancing athletic performance. [An] / [CO4]

18. Differentiate between kinetics and kinematics in biomechanics with relevant examples. [An] / [CO4]

19. Identify the center of gravity and discuss its significance in sports performance. [C] / [CO5]

20. Explain why powerlifters adjust their techniques based on joint angles and relate this to optimal force production. [U] / [CO6]

21. Define muscle contraction velocity and explain its significance in movement efficiency. [U] / [CO6]

22. Define a neuron and explain its main function. [U] / [CO5]

[2x10 = 20]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

23. Justify the role of the hamstrings in lower limb movement and their significance in sports activities. [K] / [CO1]

24. Compare concentric and eccentric contractions and their importance in weightlifting and athletic performance. [U] / [CO2]

25. Describe the key functions of bones and their relevance to strength and conditioning training. [U] / [CO2]

26. Discuss the importance of Newton's Third Law (Action and Reaction) in activities like jumping and swimming. [U] / [CO4]

27. Examine how the nervous system initiates and controls voluntary muscle actions. [C] / [CO5]

28. Discuss the force-velocity relationship and its significance in strength and power training. [U] / [CO6]

[5x4 = 20]